

Your review report

Manuscript

AirSense-R: A Multimodal Robotic Dataset for Spatially Resolved Indoor Air Quality Monitoring and Navigation Using LiDAR, IMU, and IR Camera Sensing

Feedback for the author(s)

Review file(s)

No files added.

Comments to the author(s)

The manuscript indeed contains: (1) hardware contradictions , camera version, battery count, and CO₂ sensor type (NDIR in abstract vs electrochemical MG-811 in methods); (2) asynchronous multi-modal acquisition : environmental sensors (~200ms latency) and navigation (~100ms loop) run in parallel with timestamp-based post-hoc alignment, but synchronization accuracy is unverified; (3) insufficiently specified localization, despite claiming spatially resolved data, no SLAM, odometry, or positioning algorithm is described; (4) incomplete dataset schema, no data dictionary, unclear sample counts, and undefined metadata structure.

The summary omits two critical issues: the physically impossible correlation (temp-humidity = 1.00) in Figure 15, and the complete absence of reference calibration for air quality sensors.

The manuscript structure is disorganized. Content is fragmented across too many chapters, with hardware descriptions scattered between Sections 2, 5, and 6, and Section 3 is inserted between Methods and Data Records. This poor organization makes the paper difficult to follow and obscures the core dataset description.

Confidential feedback for the Editor

Your recommendation

- *Reject*

Confidential comments to the Editor

The paper contains contradictory hardware descriptions, lacks localization methods despite claiming spatial resolution, provides no reference calibration for core sensors, and presents physically impossible validation results (temp–humidity correlation = 1.00).